

Design for Safety and Human Factors in Industrial Engineering: a review towards a unified framework

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Abstract

The rate of accidents in the industrial context is not decreasing significantly, despite the continuous effort of providing stricter safety requisites at the regulatory level. In particular, accidents' statistics show that most of accidents are related to the interaction between humans and technical systems due to a scarce consideration of human factors' issues. Based on this, the purpose of the present study is to investigate the recent research addressing design for safety (DfS) in the industrial context, focusing on those research approaches aimed at integrating human factors (HFE) within DfS activities. A systematic review was performed considering engineering journal articles that appeared in literature in the period 1995-2020. From this analysis, it emerged that most studies offered solutions intervening quite late in the design process, while much of the analyzed HFE tools should be applied during the early stages of the design process to augment their effectiveness. Additionally, a research gap between theory and practice was brought to light. Accordingly, the analysis led to the proposal of a unified design for safety process, where the main issues emerged from the literature are converged towards a holistic framework, which can support engineers in integrating human factors in DfS activities in a more practical manner.

Keywords

Design for Safety (DfS), Human Factors-Ergonomics (HFE), Occupational Health and Safety (OHS) Management, Literature Review, Human-Centered Design.

1. Introduction

In the last years standards and regulations concerning occupational safety have become increasingly strict, following the action of Governments aimed at making the framework of safety requisites to protect both consumers and professionals from unsafe products/equipment more adequately (Kim et al. 2021, Carra et al. 2019). Nevertheless, the number of accidents has not significantly decreased, especially considering small and medium sized enterprises: for instance, in Table 1 an excerpt of statistical data of occupational injuries and fatal accidents occurred in the period 2012-2019 in the European Union is reported (Eurostat 2020). In particular, a large number of accidents is connected with the use of work equipment and machinery (Gauthier and Charron 2002, Caffaro et al. 2018, de Galvez et al. 2017, Fagnoli et al. 2020), and the analysis of the reported accidents showed that most injury-causing events are related to work tasks and technical safety measures (Dzwiarek and Latala 2016). To reduce such a phenomenon, from the engineering point of view, a holistic approach for risk assessment is needed, which combines the proper design of work equipment and workplaces with the analysis of working tasks, including non-routine work and human error (Kim et al. 2021, Caffaro et al. 2017). Moreover, a risk assessment carried out at early stages of the equipment/machinery design is a key-factor for risk reduction at the lowest possible level: this is valid also in the

case of updating activities performed to augment the performances of the technical system and the overall productivity (Cordero et al. 2009). Hence, engineers have to deal with the problem of managing the interactions occurring among the product/technical system's main components, the human beings, and their environment in an effective manner (Caffaro et al. 2017, Sadeghi et al. 2016). Design for Safety (DfS) represents the most suitable solutions to such a problem, if we consider safety as a complex property of technical systems, embracing not only the safety of the system itself, but also to the safety of the operators/users that interact with system along all the phases of its life cycle (Meerkamm 1994, Sheldon 2007). Consequently, properties such as reliability, maintainability, ergonomics, and environmental concerns are strictly related to safety, impacting on its level largely (Pighini 2000).

Table 1. Accidents at work by economic activity in EU-27, 2012-2018 (adapted from: Eurostat 2020).

NACE (Section)	Non-fatal accidents							Fatal accidents						
	(thousands)							(number)						
	2012	2013	2014	2015	2016	2017	2018	2012	2013	2014	2015	2016	2017	2018
Total (all activities)	2 937.7	2 936.7	3 031.6	3 030.1	3 112.7	3 116.7	3 124.8	3 757	3 408	3 562	3 643	3 336	3 272	3 332
Agriculture, forestry and fishing (A)	143.9	151.5	169.7	162.9	161.8	148.3	144.0	485	442	507	477	482	408	441
Mining and quarrying (B)	12.0	11.4	10.0	9.2	9.0	9.5	8.4	76	69	70	70	64	43	52
Manufacturing (C)	637.5	613.0	591.7	591.9	600.9	592.0	596.5	631	563	558	632	525	472	505
Construction (F)	395.7	354.0	351.7	347.3	347.5	352.6	362.6	826	733	740	767	672	679	682
Wholesale and retail trade (G)	387.8	392.3	377.7	380.3	386.5	384.1	377.2	332	314	306	300	282	283	252
Transportation and storage (H)	251.2	254.8	241.7	245.4	242.7	272.7	280.2	556	486	570	585	562	558	557
Accommodation and food service activities (I)	146.4	149.8	144.4	150.6	158.8	159.5	163.2	40	55	70	68	56	77	53
Administrative and support service activities (N)	238.1	237.3	231.7	237.8	256.1	279.9	289.7	204	206	192	192	175	205	225
Public administration and defence (O) (*)	112.6	155.9	222.3	217.2	211.8	186.5	194.6	72	104	93	110	125	83	80

This twofold nature of safety can be addressed distinguishing in Design for Product Safety (DfPS) and Design for Human Safety (DfHS) as argued by Sadeghi et al. (2015), although these two aspects are strictly interwoven each other (Rausand and Utne 2009). The achievement of a safe human interaction with technical systems and their work environment requires a proper design and management activity, involving disciplines such as anthropometry, physiology, and biomechanics that are comprised in the field of Human Factors or Human Factors and Ergonomics (HFE), or simply Ergonomics as outlined by Karwowski (2005). Ergonomics involves the study of the organizational, environmental, cognitive, and job factors, focusing on the human and individual characteristics that impact on the workers' behavior. In recent years, the attention to the human factors has been stressed increasingly in the industrial engineering field (Muramatsu and Kato 2019). However, few studies have dealt with the problem of how to integrate safety and ergonomics in the design and management of work equipment in a practical manner (de Galvez et al. 2017). Moreover, it should be noted that also safety regulations concerning occupational health and safety (OHS), machinery safety, and safety of products to some extent require that technical systems should be undesignated and provided to users taking into account risks of accident arising from both their foreseeable use and misuse. In addition, they address manufacturers in considering during risk assessment the potential variability of the users that are going to employ the equipment during its foreseeable lifetime (European Union 2006). Due to the large variety of work equipment, machinery, products, and technical systems employed and manufactured in the industrial context, in this study these terms are used synonymously; similarly, for the terms ergonomics and human factors, indicated with acronym HFE, as explained above.

1.1 Objectives

To reduce the above-mentioned gaps, the present study proposes a review of the extant literature focusing on DfS in the industrial engineering context, with the goal of bringing to light those approaches that take into account HFE issues. The final goal of this study consists in the convergence of the analyzed design approaches towards a unified design framework.

2. Materials and Methods

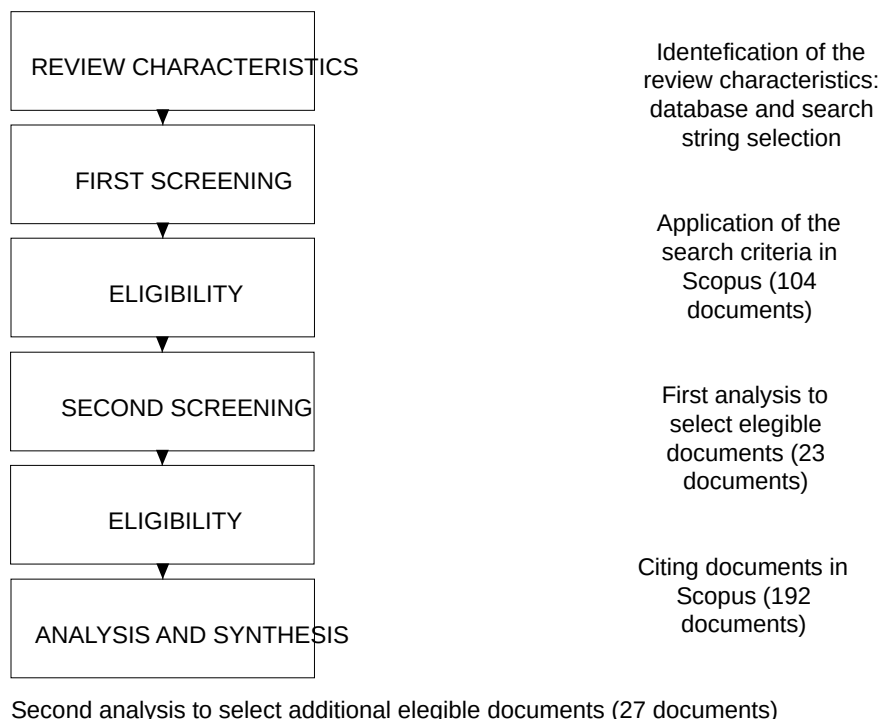
Searching in the scientific databases, it emerged that most studies addressing DfS are related to the construction sectors, where DfS is also called "Prevention through Design" or Construction Hazard Prevention through Design (CHPtD) (Poghossyan 2018, Hardison and Hallowell 2019). Actually, the concept of PtD is broader since NIOSH defines it as follows: "PtD encompasses all of the efforts to anticipate and design out hazards to workers in facilities, work methods and operations, processes, equipment, tools, products, new technologies, and the organization of work" (NIOSH 2021). Narrowing the analysis, data collection was based on the selection of articles published in

engineering journals from 1995 to 2020 in the Scopus database. More in detail, a systematic literature review was carried out following the PRISMA approach (Thomè et al. 2016), which consists in the following steps:

- Identification of material (definition of the scope, database, and search criteria);
- Screening of pertinent articles (application of the search criteria);
- Eligibility evaluation (abstract analysis for inclusion/exclusion);
- Data analysis and synthesis (full-text review).

As far as the search criteria are concerned, the search string “design for safety” in the title, abstract and keywords was used to search journal articles published in English in the subject area “Engineering” in the period 1995-2020. As a result, 104 were found, which were screened to select those related to the field the design of safe products such as machinery, machinery components and work equipment, including those facing design for safety problems of complex systems, such as electromechanical components, as well as studies addressing interventions to improve the safety performances at industrial work sites. Conversely, articles focusing on problems related to safety in other sectors such as transportation (e.g. road safety), construction (e.g. structural safety, design stages of construction works), naval engineering (ship design), electronics, nuclear plants, software engineering, etc. were excluded, while studies on the design of work equipment used in the latter sectors as well as in other activities were considered in the review. The result of this first screening consisted in the selection of 23 articles. Then, to make the analysis broader, the citing documents of these studies reflecting in Scopus were taken into account as well. The output of this secondary source consisted in 192 additional documents belonging to different subject areas, which were analyzed using the above screening criteria. The result of this analysis provided 27 further articles (excluding duplicates), which were included in the review (in Figure 1 the whole selection process is summarized). Data analysis and synthesis were then carried out by going through the full text of the 50 selected documents. The next step consisted in bringing to light the research trends emerging from these studies. This analysis was aimed at investigating the research trends, i.e. the emerging themes that academics and researchers have dealt with in the industrial engineering DfS context. With this goal in mind, as classification benchmarks, the following research themes were used:

1. Skills and knowledge of design for safety engineers (SKILLS);
2. Re-engineering studies aimed at improving the safety level of products/equipment (REENG.);
3. Design of production sites to augment workplace OHS (WORKPL.);
4. Theoretical frameworks development for the safety improvement of specific technical systems or the enhancement of their properties that are strictly related to safety (FRAMEW.); and 5. Methods proposing a novel design for safety approach (METHOD).



Classification and synthesis of the selected 50 documents

Figure 1. Scheme of the literature review process.

3. Results

The distribution of the selected studies in the period 1995-2020 is reported in Figure 2 (no. of publications per year), while their full list organized per the research themes outlined in the previous section is reported in Table 2. As summarized in Figure 3, the majority of articles focused on the development of theoretical frameworks, while a few addressed workplace DfS approaches and safety engineers' skills.

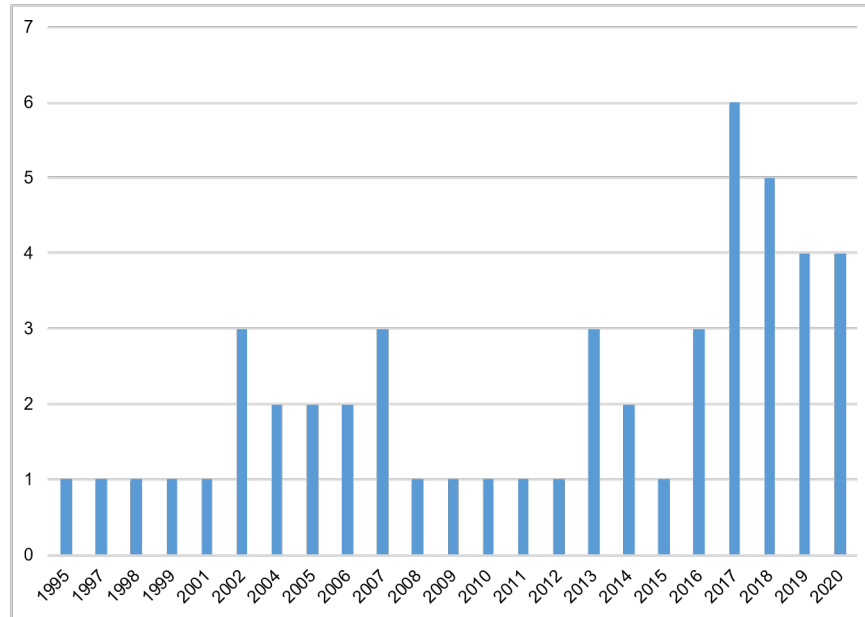


Figure 2. No. of publications per year.

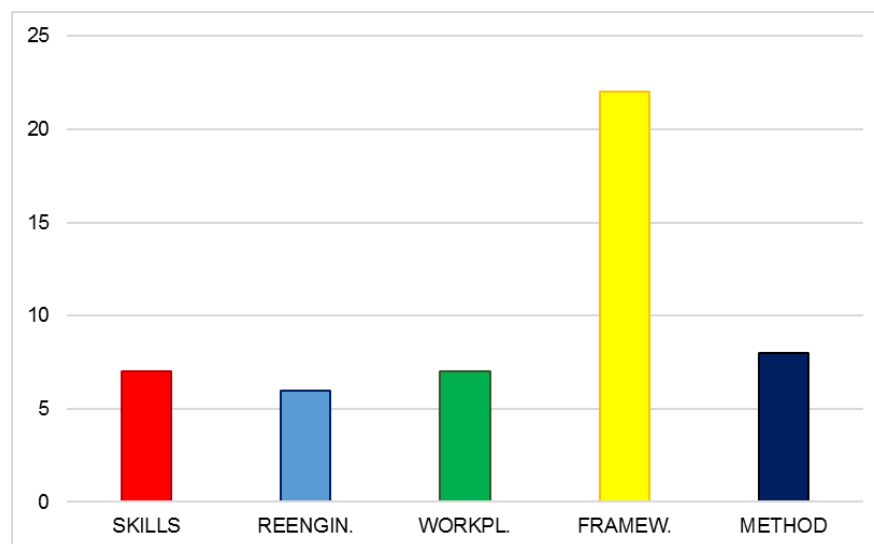


Figure 3. no. of publications per each research theme.

3.1 Skills and Knowledge

Studies belonging to this group focused on two different issues:

1. the skills and knowledge a safety engineer should acquire;
2. the safety knowledge, or safety perception related to the equipment user.

Regarding the first point, it is noteworthy mentioning the DfS competencies outlined by Gauthier and Benoit (2002) in order to ensure a better integration of safety from the start of the design process, which include: the use of an efficient design methodology; the analysis of the users' needs; the application of safety technologies; knowledge of risk management; team-working attitudes; communication skills; ethics and professionalism; and promotion of such approach. In this context, HFE issues are considered within the analysis of the users' needs and expectation, suggesting the use of task analysis, focus groups, and interviews as means to adapt the technical system to users to allow a higher level of safety, functionality and performance. The involvement of the users is also at the basis of the study by van Duijne et al. (2007), who proposed such an approach to augment the skills and knowledge of engineering students through a redesign project aimed at augmenting the safety level of a gas lamp. De La Garza and Fadier (2005), investigating the activity of different types of engineers (experienced, non-specialized engineers, and draftsmen) and their different approach to safety issues, observed that usually design integration of safety and ergonomics is not included in the initial phases of the design process of industrial equipment, since engineers pay more attention to standards' and regulations' requirements. For these reasons they suggest a proactive approach entailing both industrial and individual variability to assure that both the risks associated with the equipment technical parts and those linked to the users' health and safety are taken into account effectively.

Similarly, Hale et al. (2007) stressed on the fact that engineers need to apply two complementary approaches to safety in design: the human-centered approach and the barrier approach. Accordingly, to reduce such a "mental workload" on designers, it is suggested the use of methods of risk analyses that can support them in dealing with both the physical and behavioral aspects, to include operational inputs in the design activity.

Regarding the safety perception related to the equipment user, Suhanyiova et al. (2020) observed that to better understand the concept of "safety system" company training related to product safety behaviors (e.g. tool handling, safety protocols, awareness courses) is needed to assure a proper product design, assembly, servicing and maintenance. In addition, Zhou et al. 2016 investigating the relationships between DfS and Management Commitment to safety, outlined that management commitment to safety, product safety culture, and DfS are significant product safety predictors.

Table 2. List of the selected studies (in alphabetical order).

Authors	Year	Type	Authors	Year	Type
Ahmad et al.	2019	WORKPL.	Moraes et al.	2014	REENG.
Baklouti et al.	2020	FRAMEW.	Nissanke and Dammag	2002	WORK
Benabdellah et al.	2019	FRAMEW.	<u>Orsoni</u>	2006	WORK
Caputo et al.	2013	METHOD	Park and Park	2004	FRAMEW.
Cecchini et al.	2017	REENG.	Pinto et al.	2013	SKILLS
Chang and Cheng	2010	FRAMEW.	Qi et al.	2008	METHOD
Compare et al.	2018	REENG.	Ramírez-Marengo et al.	2013	FRAMEW.
Dantan et al.	2019	WORKPL.	Rausand and Utne	2009	FRAMEW.
De la Garza and Fadier	2005	SKILLS	Sadeghi L. et al.	2017	METHOD
Debouk and Wysocki	2005	FRAMEW.	Sadeghi, L. et al-	2015	METHOD
Dowlathshahi	2001	FRAMEW.	Sadeghi, L. et al.	2016	FRAMEW.
Eddy et al.	2017	REENG.	Sadeghi, S. et al.	2017	METHOD
Fargnoli and Lombardi	2019	METHOD	Schupp et al.	2006	WORK

Fargnoli and Sakao	2017	FRAMEW.	Sen et al.	1999	FRAMEW.
Fargnoli et al.	2018	METHOD	Sheldon	2004	FRAMEW.
Fargnoli et al.	2014	FRAMEW.	Silva et al.	2016	FRAMEW.
Gauthier and Charron	2002	REENG.	Silvennoinen et al.	2017	METHOD
Gauthier, F., Benoît, R.	2002	SKILLS	Suhanyiova et al.	2020	SKILLS
Guini et al.	2018	FRAMEW.	van Duijne et al.	2007	SKILLS
Hale et al.	2007	SKILLS	<u>Verma et al.</u>	2020	WORK
Houssin and Coulibaly	2011	WORK	Wang and Ruxton	1998	FRAMEW.
Jeang et al.	2018	FRAMEW.	Wang and Ruxton	1997	FRAMEW.
Kim and Zuo	2018	FRAMEW.	Wang et al.	1995	FRAMEW.
Kongwat et al.	2020	REENG.	Wysocki, and Debouk,	2007	FRAMEW.
Liscouët et al.	2012	FRAMEW.	Zhu et al.	2016	SKILLS

3.2 Re-engineering

In this category, articles addressing the upgrade of technical systems and processes in order to augment their safety level and usability (e.g. updating them to mandatory safety requirements) were listed. For example, Cecchini et al. (2018) proposed a decoupling system, to be mounted on a manure spreader. Such a solution is aimed at being automatically activated when the wheels of the wagon are not moving to prevent the contact with the moving rotating parts of the machine when the worker is inside the hopper (which is a behavior not allowed by safety regulations but quite common among farmers). Other articles have proposed the redesign of technical systems and processes based on the users' feedback. In this context, it is worth mentioning the study by Moraes et al. (2014), who proposed a production process redesign based on the analysis of the variability emerged from mapping the workers' activities, including their actions, visualization and communication needs and features. The variability analysis also concerned the identification of problems experienced by workers and the risks involved in the work settings. Differently, Compare et al. (2018) proposed an approach based on the machinery safety standards to update workplace production equipment: in the risk assessment activity, specific concerns are given to the information delivered to the users of the equipment, such as pictograms, visual-audible alarms, as well as training on working and emergency procedures.

3.3 Design of production sites

Several articles have dealt with OHS problems at the workplace, providing design solutions to improve safety and ergonomics. In particular, Orsoni (2007) proposed a tool for the implementation of DfS criteria during plant design, involving handling, processing, and/or storage of hazardous materials. Such an approach is based on fuzzy logic and simulation tools to capture concurrent hazards combination and provide overall risk assessment. Another supportive tool for risk assessment is proposed by Dantan et al. (2019), who introduced a design framework based on the function-behavior-structure to take into account human factors in the design process. Such an approach allows the analysis of the behavioral interactions between workers and manufacturing equipment generated by the system structure: in fact, modelling equipment and human interaction provides information on how fatigue and other factors can impact the work activities, reducing the probability of errors and accidents. Differently, Nissanke and Dammage (2002) proposed a software tool (named Safecharts) to classify systems' transitions according to the nature of risks they carry and extend the specification of each transition with additional guards and enforcement conditions. Accordingly, this tool allows engineers to better estimate the safety-oriented definition of default states of a certain technical system. Schupp et al. (2006) proposed a software tool based on a collaborative Technology Management Environment (TME) and the Safety Modelling Language (SML) with the aim of providing a support for gathering information about hazards and augmenting communication between designers and operators about how the risk

control has been designed and what factors are crucial to its continuing effectiveness. Finally, Houssin and Coulibaly (2011) proposed an approach to model working situations and solve risk situations and contradictions.

3.4 Frameworks

The majority of the selected articles proposed procedures or methodologies to improve safety by means of the combined use of different design tools. In such a context we can distinguish research providing design methodologies and the combined use of different tools to solve specific safety problems, as well as articles analyzing and discussing different DfS approaches of a more general nature. More in detail, most combined tools concern advanced solutions to augment the effectiveness of specific well-known safety tools by means of supportive techniques, such as fuzzy logic, Boolean Representation Method (BRM), or the Analytic Hierarchy Process (AHP). For example, Wang et al. (1995) proposed an inductive bottom-up risk identification and risk estimation framework incorporating the Failure Mode, Effects and Criticality Analysis (FMECA) and BRM tools to define the system failure events and the associated causes, especially when multiple state variables and feedback loops are generated. Similarly, Sen et al. (1999) proposed a methodology combining probabilistic risk analysis tools such as FMECA and BRM model, to assess and analyze the top events and their causes. This information is linked directly for risk Sensitivity Analysis (SA) and Multiple Objective Decision Making (MODM) based reliability optimization. Another example is given by Baklouti et al. (2020), who proposed a framework to generate Dynamic Fault Trees (DFTs) of technical systems that contain redundancies, which is widely used in safety-critical systems to enhance their reliability. This type of studies, however, do not specifically address HSE problems, mainly focusing on other properties related to safety, such as reliability and availability. Differently, Rausand and Utne (2009) proposed a methodology consisting in 8 different phases, which takes into account the whole lifecycle of product from the manufacturer's standpoint. In each one of these phases, safety-related issues are provided ranging from the analysis of safety requirements in laws, regulations, and standards at the beginning of the process (planning phase). The involvement of the users is also foreseen to ensure customer satisfaction and avoid the possible misuse of the product. Fagnoli et al. (2014) proposed a methodology for the sustainable development of industrial products. Such an approach includes not only environmental aspects, but also social issues, which from the manufacturers' point of view can be translated into a deeper attention to the product's impact on the users, stressing on safety and ergonomics features of the product. A more comprehensive approach is proposed by Sadeghi et al. (2016), who focused on the concept of human safety, which concerns accident prevention in work situations (i.e. the interaction between the operator, the work equipment, and the work environment). Other studies have analyzed DfS approaches in the ambit of DfX strategies (where the "X" represents a general property of the system). In this ambit, Benabdellah et al. (2019), analyzing the extant literature on DfX tools, raised the problem of how to consider the variability of main components, human beings, equipment, and their environment at the design stage, fostering the use of tools aimed at merging the human factors to the safety dimensions of a product. Differently, Fagnoli and Sakao investigated the application of the Quality Function Deployment (QFD) method in DfX research, testing the effectiveness and usability of QFD based tools aimed at the improvement of the safety level, considering both the manufacturers' perspective (Safety Function Development) and the users' point of view (Hazard Function Deployment).

3.5 Methods

In this group, articles proposing a novel design for safety approach issues were considered. Focusing on those considering human factors, it is worth mentioning the work of Sadeghi L. et al. (2017), who proposed the use of the Function–Behavior–Structure (FBS) modeling integrated by the Product-Service System (PSS) approach in order to consider product behaviors and its interaction with service activities in the work situation analysis. Differently, Fagnoli et al. (2018) focused on the analysis of work activities related to the use of an equipment, proposing a risk assessment methodology based on the integrated use of QFD with fuzzy logic and the Delphi method, in order to take into account operators' practical concerns regarding the equipment usage problems during the different work activities they are involved in. Instead, Sadeghi S. et al. (2017) developed a software tool to make the recording and analysis of accident related to the use of a work equipment more efficient: in particular, a specific attention was paid to the collection of information concerning the behavior and capabilities of the operator. Fagnoli and Lombardi (2019) proposed a bottom-up approach for the analysis of human safety performances to investigate the perception of hazards and risk-taking behaviors of workers. The inclusion of human reliability tools in the safety assessment of working activities showed the possibility for safety engineers to bring to light the underestimation of potential risks by equipment operators. Besides, Caputo et al. (2013) proposed a tool for the selection of safety devices of work

equipment based on the machinery safety standards: in this context, specific selection criteria were provided to deal with the problem of safety device misuse not only during production activities, but also during the machinery operations, such as adjusting and cleaning. For this purpose, the assessment of the following 15 factors useful to estimate the suitability of safety devices is suggested: Effectiveness (E), Protection from falling objects (PFO), Cost (C), Reliability (R), Maintainability (M), Fault tolerance (FT), Detectability (DET), Interference avoidance (IA), Tampering avoidance (TA), New hazards avoidance (HA), Flexibility (FL), Ease of use (EU), Useful life (UL), Ease of installation (EI), Safe lubrication (SL).

3.6 Development of a unified approach

From the analysis carried out a maturity model integrating DfS and HFE tools can be outlined, i.e. a unified approach merging the salient research trends emerged from the extant literature. This framework mainly concerns the first stages of the design process, where main decisions are taken, i.e. planning, conceptual design, and embodiment design. However, detailed design, prototyping and product launch steps also deem attention, since a more complete modelling of the work situations (i.e. the combination of the technical system with its users and the workplace) can provide further information on safety and ergonomic issues. The whole procedure is reported in Figure 4, where for each phase the main activities and outputs are indicated, together with the tools and methods emerged from the literature review. In detail, in the planning phase, when deciding the general requisites of the technical system, most relevant activities from the safety and ergonomics point of view should concern the definition of the users' needs and the analysis of accidents related to similar systems in order to collect information about the users' behavior and the possible misuse of the system. Then, in the second phase, where the conceptual design of the system is defined, it is possible to apply more specific tools to analyze the human safety of the system. In such a context, the analysis of working tasks and situations is essential not only to understand human behavior, but also to avoid conflicts that might be generated by the presence of other equipment in the workplace. In the embodiment design phase, the information about the system is sufficient to apply not only qualitative tools, but also quantitative methods for risk analysis. Hence, also more specific HFE methods can be used, to evaluate hazards due to noise and vibrations, as well as to evaluate the proper safety devices. Moreover, at this level of definition of the system information concerning its lifecycle is more precise:

accordingly, installation and maintenance operations can be included in the risk assessment activities.

PHASE	ACTIVITIES	OUTPUT	DfS and HFE TOOLS
PLANNING	• Analysis of mandatory requirements (Legislation, contracts, etc.) • Analysis of stakeholders' requirements (Manufacturer/Users/Operators' needs, maintenance service, etc.) • Analysis of the technical system's life-cycle and possible scenarios • Analysis of accidents' and incidents' reports and statistics	• Main requisites of the system • Feasibility analysis (production, installation, maintenance, disposal, etc.)	• Preliminary Hazard Analysis (PHA) • Interviews with users • Accidents' reports • Safety Function Deployment (SFD) • Axiomatic Design (AD)
CONCEPTUAL DESIGN	• Analysis of the system's functions and sub-functions • Evaluation of possible conceptual solutions • Preliminary risk assessment analysis • Life-cycle scenarios assessment • Human Reliability Analysis	• Conceptual model of the system	• Preliminary Human Safety Analysis (PHSA) • Hazard Function Deployment (HFD) • Function Behavior Structure (FBS) • Failure Modes and Effects Analysis (FMEA) • Fault Tree Analysis (FTA) • Variability Analysis • Safecharts
EMBODIMENT DESIGN	• Identification of the main features of the system (dimensions, materials, life-cycle, etc.) • Conformity assessment (EHRSR, OHS, etc.) • Work situation analysis • Subsystem Hazard Identification • Subsystems Failure analysis	• Definition of the systems' main features • Definition of the systems' safety issues	• Computer musculoskeletal modeling (CMM) • Multi-Objective Optimization (MOO) • Failure Modes Effects and Criticality Analysis (FMECA) • Fault Tree Analysis (FTA) • Event Tree Analysis (ETA) • Semantic Distances Analysis • Function Behavior Structure (FBS) • Hazard and Operability analysis (HAZOP) • Safecharts • Safety Devices Evaluation

Figure 4. General scheme of the unified framework integrating HFE in DfS activities.

4. Discussion

The study carried out aimed at providing an answer to the following research question: “Which is the path safety engineers have to follow in order to harmonize technical safety aspects with all the concerns associated with the relationship user-technical system in an industrial context?”. Needless to say, from the analysis of the extant literature there is not a unique response due to the large variability of the work contexts and the different types of equipment/technical systems used and manufactured. However, focusing the analysis of the human-centered DfS approaches, some findings can be outlined.

The majority of the selected articles proposed theoretical approaches that are based on the augmentation and combination of commonly used DfS tools by means of decision-making techniques (mostly addressing hazard and failure analyses or the improvement of reliability and availability of the systems). Conversely, a few studies dealt with the analysis of workplaces, although such aspect plays a fundamental role in the interaction user-technical system. It also noteworthy to mention the lack a specific knowledge among engineers about safety and ergonomic issues. Studies addressing this aspect have shown that although engineers are aware of these aspects, they hardly know how to put them into practice in design and development activities. In other words, it emerged a gap between “work as imagined” and “work as done”, i.e. the daily work situations and events. Based on this, the proposition of methodologies illustrating the proper use of tools integrating DfS and HSE is recommended largely with the goal of contributing to bridge the gap between theory and real-world practice.

Investigating such aspect more in detail, it emerged that the literature reviewed indicate that most studies offered solutions intervening quite late in the design process, mainly in the detailed design phase, when most relevant decisions on the product/process features are already taken. However, to augment their effectiveness, much of the analyzed tools should be applied during the early stages of the design process both in the case of a product development and in that related to the workplace/process design. These findings reflect the research insights by Sadeghi et al. (2017) as well as Urbanic and El Maraghy (2009) and could be used as a baseline for more detailed and larger investigation on the integration of HFS tools in the first stages of DfS activities.

Interestingly, it also emerged that HFE issues are mostly considered just as the evaluation of physical aspects related to work activities and workplace features. Unlike, psychosocial/behavioral factors are rarely included in the analysis, although to some extent mandatory safety requirements (e.g. the “Machinery Directive” (European Union 2006)) require to take into account any reasonably foreseeable misuse or abnormal situation of the equipment (Fargnoli et al. 2014, Sadeghi et al. 2015). Moreover, human factors consideration allows for a more proactive approach to safety, reducing the possibility of migrating towards an “illegal-normal” organizational functioning (Rasmussen 1997). Additionally, it has to be stressed the fact that the inclusion of tools for human factors’ analysis in the safety assessment of working activities can allow safety engineers to bring to light the underestimation of potential risks by equipment operators, reducing the gap between “work as done” and “work as imagined”.

Overall, it should be also noted that the safety management of technical systems and work equipment is poorly addressed, although the need to consider the whole life-cycle of them is suggested (Rausand and Utne 2009). As a matter of facts, a life-cycle perspective should take into account the various activities characterizing the operational phase of the systems (i.e. installation, use, maintenance, and disposal) in a more holistic approach, in line with research hints of de Galvez et al. (2017). Moreover, when carrying out risk assessment activities, a top-down approach should be merged with a bottom-up procedure that integrates the analysis of work situations, safety climate and risk perception of workers, e.g. investigating the users’ needs and their feedback (Pinto et al. 2013, Fargnoli et al. 2012, Falzon 2008).

Hence, the proposed procedure can represent a possible answer the research question raised above, summarizing a human-centered design for safety unified framework, which can support engineers in reducing the gap between theory and practice. Although this research work cannot be considered exhaustive, and the described procedure might appear too simple solution to provide a precise means to augment the design and management of work equipment and technical systems, it undoubtedly represents a powerful approach to bring to light both safety and ergonomics features of work situations in an industrial context, augmenting knowledge in this emerging research domain.

5. Conclusion

The study proposed a review of the extant literature concerning DfS in the industrial context, focusing on those studies that dealt with human factors’ issues. Overall, the contribution of the study does not rely only on providing a general overview on the status quo of DfS applications and tools to improve health and safety of workers/users dealing with technical systems, but also on the identification of major issues dissociate to the consideration of ergonomics in the design and management of such systems. Based on this, a general framework unifying the salient issues and methods towards a human-centered design for safety approach was developed, as a result of a comprehensive research synthesis and analysis, which can certainly contribute to augmenting knowledge in the safety engineering domain. A possible extension of this study for future work would be the inclusion of a larger number of tools as well as an augmented methodical approach taking into account different subject areas.

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